JOC The Journal of Organic Chemistry

pubs.acs.org/joc Note

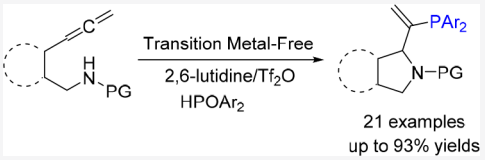
Metal-Free Intramolecular Aminophosphination of Allenes

Zhantao Yang, Zhenzhen Sun, Ningning Zhu, Wenyan Li, Meng Han, Yan Qiao, Chun-Hua Yang,* and Junbiao Chang*

Cite This: J. Org. Chem. 2020, 85, 15686−15692 Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

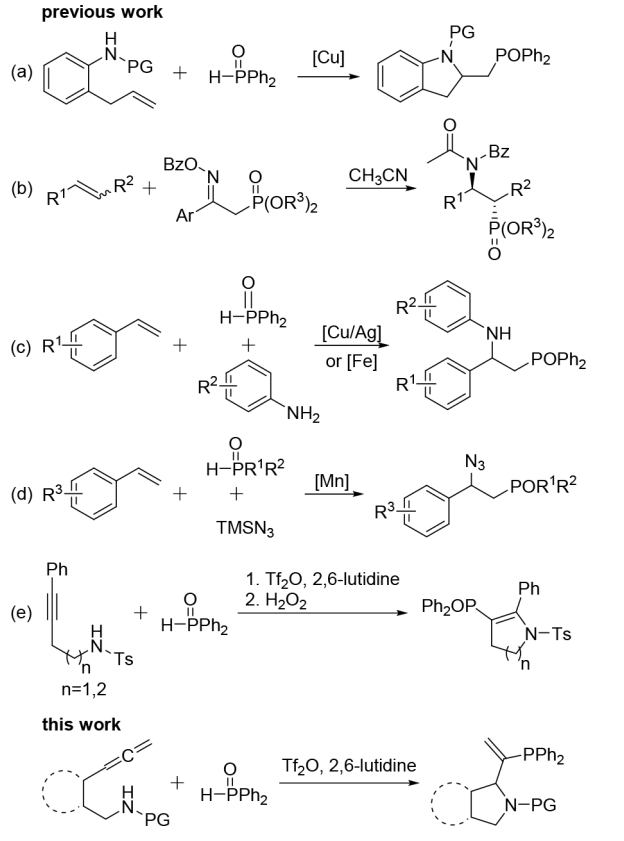
ABSTRACT: A metal-free intramolecular aminophosphination of sulfonamidoallenes with diarylphosphine oxides and Tf_2O was developed. This method offers a general and practical procedure to construct valuable alkenylphosphine-substituted N-heterocycles via the bifunctionalization reaction of allenes in good yields under mild conditions.



Downloaded via FUDAN UNIV on February 20, 2021 at 02:52:26 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

Organophosphorus compounds have drawn tremendous attention due to the widespread applications as building blocks of biological molecular,[1] drug and drug candidates,[2] and ligands[3] in organic synthesis. In particular, β-aminophosphines, bearing unique properties in the field of coordination chemistry and asymmetric catalysis because of their broad electronic and steric tenability, have gained considerable attention.[4] Furthermore, β-aminophosphines are privileged structures in drug development, for example, antimalarial,[5] antiviral,[6] and antibacterial[7] activities. To this end, significant efforts have been made toward the preparation of β-aminophosphines or β-aminophosphonates. The construction of C−P bonds via the C−C multiple bond bifunctionalization reaction catalyzed by a transition metal was one of the most efficient methods. However, this method is usually not feasible because the transition metal catalyst forms the coordinatively saturated complex with PR_2H or PR_2R', and the catalytic activity is reduced or completely lost.[8] Therefore, examples that are involved in the construction of C−P bonds via C−C multiple bond bifunctionalization reactions are still limited. The Yang group reported an efficient method for the synthesis of various phosphorated indoles by the cyclization of Ts-protected *o*-alkenylanilines catalyzed by Cu (II) in high yield (Scheme 1a).[9] Then, they reported a method to synthesize β-aminophosphonates with oxime phosphonates (Scheme 1b).[10] Zhang et al. discovered a novel intermolecular aminophosphinoylation using $AgNO_3$ to produce a phosphinoyl radical from diarylphosphine oxides, followed by amination with $CuBr_2$ as a catalyst to obtain β-aminophosphonates (Scheme 1c).[11] Fang et al. also reported a new alkene bifunctionalization reaction to gain β-aminophosphonates with anilines and H-phosphonates using $FeCl_3$ through a radical process (Scheme 1c).[12] Tang and co-workers developed a new and simple method to prepare β-azidophosphonates by $Mn(OAc)_3$-catalyzed oxidative phosphonation azidation of olefins (Scheme 1d).[13] However, almost all of these methods suffer from the requirement of transition metals, and external oxidants are usually needed. These requirements have limited the substrate scope and the practical utility. Therefore, it is still

Scheme 1. Aminophosphination/Aminophosphinoylation of C−C Multiple Bonds



Received: September 8, 2020
Published: October 29, 2020

ACS Publications © 2020 American Chemical Society

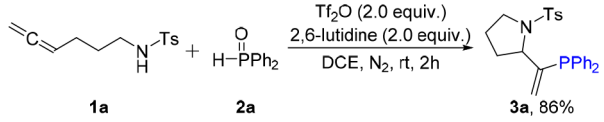
https://dx.doi.org/10.1021/acs.joc.0c02169
J. Org. Chem. 2020, 85, 15686−15692

desirable and essential to develop environmentally benign and metal-free methods for the direct preparation of β-aminophosphine derivatives.

Recently, metal-free C−heteroatom bonding as an environmentally friendly process has garnered interest from the synthetic community.[14] These methods are very beneficial as promising substitutes and valuable supplements for transition-metal catalytic processes. In 2017, Hirano and Miura et al. reported a metal-free electrophilic phosphine cyclization of alkynes.[15] In the presence of Tf_2O/2,6-lutidine, the electrophilic phosphination reagents, which are produced in situ by diarylphosphonate oxides, are successfully coupled with alkynes to give 2-phenyl-substituted β-aminophosphination products (Scheme 1e).[15a] However, to the best of our knowledge, the current studies of aminophosphination/aminophosphinoylation mainly focus on alkenes and alkynes. It is a challenge to obtain vinylphosphine-substituted N-heterocycles using conventional methods. Thus, it is highly desirable to develop a general method for the aminophosphination of allenes and to provide a general method for the preparation of alkenylphosphines, which could be used as viable substrates in subsequent organic reactions.[16] As part of our development of an approach for the formation of C−P bonds,[17] we have, for the first time, disclosed a metal-free aminophosphination of allenes toward the structurally diverse alkenylphosphines via the bifunctionalization reaction of allenes concomitant with the formation of the C−N and C−P bonds. This metal-free strategy makes it easy and efficient to obtain complex alkenylphosphine-substituted N-heterocycles, which are of great significance in medical chemistry.

We initially selected *N*-(hexa-4,5-dien-1-yl)-4-methylbenzenesulfonamide **1a** and diphenylphosphine oxide **2a** as model substrates and tested various Ag and Cu catalysts in different solvents at different temperatures. Unfortunately, no phosphinative cyclization products were obtained. Inspired by Hirano and Miura's result[15a] and our previous success,[17] we carried out the aminophosphination using 2,6-lutidine and Tf_2O. The reaction of **1a** (0.2 mmol) with **2a** (0.4 mmol) in the presence of 2,6-lutidine (0.4 mmol) and Tf_2O (0.4 mmol) was investigated in 1,2-dichloroethane (2 mL) for 2 h at room temperature. To our delight, the phosphinative cyclization product **3a** was obtained in 86% yield (Scheme 2).

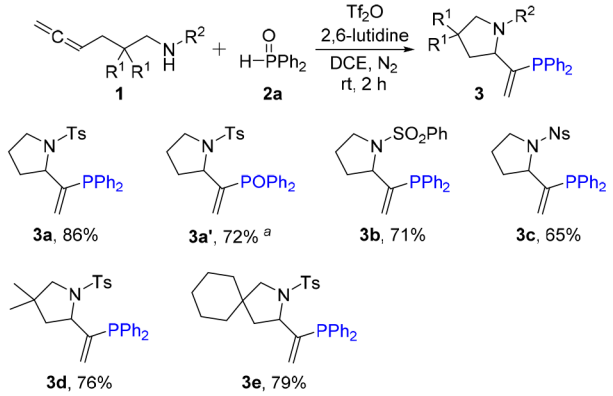
Scheme 2. Intramolecular Aminophosphination of 1a



Encouraged by this preliminary result, the scope of the substrates was studied. The results are summarized in Scheme 3. First, an investigation into a series of N-protecting groups showed that sulfonyl groups (**1a−c**) provided the corresponding products in 65−86% yields. The aminophosphinoylation product **3a′** could be obtained (72%) when the reaction mixture was treated with H_2O_2. Moreover, different *gem*-disubstituted substrates (**1d−e**) could still afford the corresponding products in 76−79% yields, indicating that the reaction was not affected significantly by the Thorpe−Ingold effect.

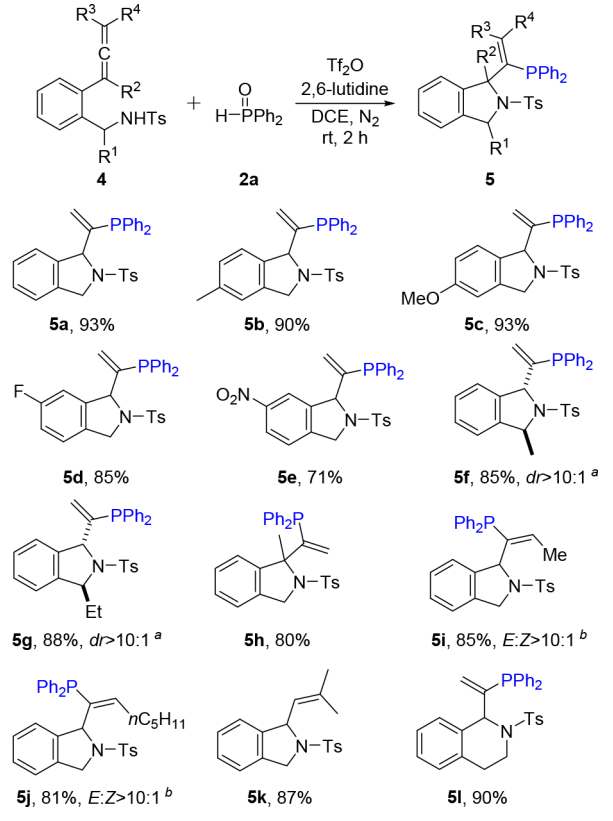
Next, 2-allenylphenylmethanamine substrates were subjected to the study to further expand the scope of the reaction and to get important functionalized isoindolines. The results are summarized in Scheme 4. 2-Allenylphenylmethanamines

Scheme 3. Scope of Intramolecular Aminophosphination of Allenes



[a] The reaction mixture was treated with H_2O_2.

Scheme 4. Scope of Intramolecular Aminophosphination of 2-Allenylphenylmethanamine



[a]The dr values were calculated through the NMR spectrum of the crude product mixture. [b]The *E*/*Z* values were calculated through the NMR spectrum of the crude product mixture.

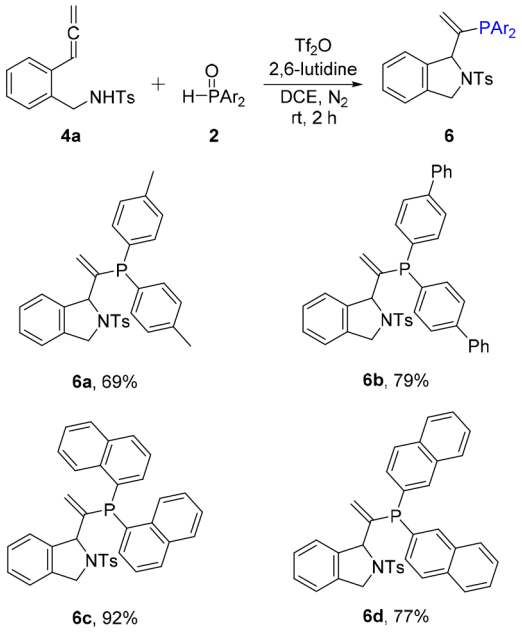
bearing electron-neutral (**4a,b**) or electron-donating groups (**4c**) on the aryl ring were well tolerated, providing the desired products in 90−93% yields, whereas electron-withdrawing groups such as F (**4d**) and NO_2 (**4e**) decreased the yield. In addition, a stereogenic center at the α-position of the nitrogen atom was introduced (**4f,g**), and the expected products were obtained in 85−88% yields. Configuration of the product 1,3-

https://dx.doi.org/10.1021/acs.joc.0c02169
J. Org. Chem. 2020, 85, 15686−15692

anti-**5g** was established by single-crystal X-ray diffraction experiments. Density functional theory (DFT) calculations indicated that the energy of the antitransition state ($TS1^{anti}$) was lower than the syn-transition state ($TS1^{syn}$), due to the advantageous C−H···π interactions between the ethyl group and the two phenyl groups in $TS1^{anti}$, which is absent in $TS1^{syn}$ (see Supporting Information for details). For the 1,1-disubstituted allene **4h**, the reaction proceeded successfully, giving the desired product in 80% yield. When the substrates with an alkyl group at the terminal allene (**4i**−**j**) were subjected to the reaction, the corresponding products (**5i**−**j**) were obtained in good yields. Configurations of the product *E*-**5i** were established by the NOESY spectrum experiments. We also examined the substrate with a trisubstituted allene (**4k**). Disappointingly, only the hydroamination product (**5k**) was obtained, which might primarily result from the increased steric hindrance of methyl groups. Moreover, the reaction could also proceed well by using *N*-(2-(buta-2,3-dien-2-yl)benzyl)-4-methylbenzenesulfonamide (**4l**) and afford the corresponding product (**5l**) in 90% yield.

Subsequently, several phosphine oxides **2** participated in the aminophosphination reaction with **4a**, and the results were illustrated in Scheme 5. Di-4-tolylphosphine oxide and

Scheme 5. Scope of Phosphine Oxides 2

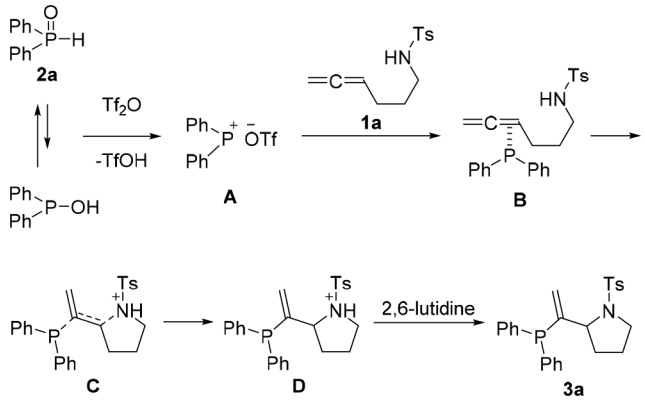


di([1,1′-biphenyl]-4-yl)phosphine oxide provided the desired vinyl phosphine **6a**,**b** in moderate to good yields. Sterically, di(naphthyl)phosphine oxide furnished the desired product (**6c**,**d**) in a good yield. We have tried other phosphine sources, such as HP(O)Ph(OEt), dibenzo[*c*,*e*][1,2]oxaphosphinine 6-oxide. However, only trace aminophosphination product was detected.

Density functional theory (DFT) calculations indicated that the aminophosphination of sulfonamidoallene undergoes bifunctionalization of allenes to form the C−N and C−P bond (see Supporting Information for details).

On the basis of literature[15a] and DFT calculated results, a tentative mechanism is proposed and shown in Scheme 6. The electrophilic phosphorus species **A** was generated from **2a** and Tf_2O, which was coordinated to allene **1a**, giving the allene-

Scheme 6. Tentative Reaction Mechanism



phosphine complex **B**. Nucleophilic attack on the allene-phosphine complex **B**, along with the C−P bond formation, could afford intermediate **C**, which rapidly converted to the corresponding alkenylphosphine intermediate **D**. Deprotonation of intermediate **D** in the presence of 2,6-lutidine would lead to the formation of the vinylphosphine-substituted pyrrolidine product **3a**.

In summary, we have successfully developed the first efficient, metal-free intramolecular aminophosphination of various allenes with diarylphosphine oxides, affording a rapid strategy for a structurally diverse array of alkenylphosphine-substituted N-heterocycle bifunctionalization reactions of allenes concomitant with the formation of the C−N and C−P bonds under mild conditions. Importantly, the atom economy and operational simplicity related to this method suggest its great potential for broad applications in the construction of valuable alkenylphosphine-substituted N-heterocycle frameworks in medical chemistry. Moreover, DFT calculations suggest that the sulfonamidoallene undergoes a bifunctionalization reaction along with the formation of the C−N and C−P bonds with the in situ generated electrophilic phosphination reagent.

EXPERIMENTAL SECTION

Unless otherwise noted, all reagents were purchased from commercial suppliers and used without further purification. Analytical thin-layer chromatography (TLC) was performed on silica gel GF_{254}. Visualization was accomplished by irradiation with UV light at 254 nm or $KMnO_4$ stain solution. Column chromatography was performed on silica gel (200−300 mesh). ^1H NMR spectra were recorded on a Bruker DRX-400 spectrometer (400 MHz). Chemical shifts were reported in parts per million (ppm) referenced to 0.0 ppm for tetramethylsilane. Data are reported as follows: chemical shift, multiplicity (s = singlet, d = doublet, t = triplet, q = quartet, m = multiplet, br = broad), coupling constants (Hz), and integration. $^{13}C\{^1H\}$ NMR spectra were recorded on a Bruker DRX-400 spectrometer (100 MHz) and were fully decoupled by broadband proton decoupling. Chemical shifts were reported in parts per million (ppm) referenced to 77.16 ppm for $CDCl_3$. ^{31}P NMR spectra were recorded on a Bruker DRX-400 spectrometer (162 MHz). High-resolution mass spectra (HRMS) were recorded on a Waters LCT PremierxeTM (USA).

The allene substrates were prepared according to the general procedure of our previous work.[18]

General Procedures. In a Schlenk tube, allene substrates (0.2 mmol, 1.0 equiv) and phosphine oxides **2** (0.4 mmol, 2.0 equiv) were placed with a magnetic stir bar under a N_2 atmosphere. 1,2-Dichloroethane (2 mL, 0.1 M) and 2,6-lutidine (47 μL, 0.4 mmol) were added by a syringe sequentially. The resulting mixture was

https://dx.doi.org/10.1021/acs.joc.0c02169
J. Org. Chem. 2020, 85, 15686−15692

stirred until it became a homogeneous solution. Tf_2O (67 μL, 0.4 mmol) was then added by a syringe. The reaction mixture was stirred for 2 h at room temperature. Then the reaction mixture was poured into sat. $NaHCO_3$ and extracted with CH_2Cl_2. The combined organic layer was dried over anhydrous Na_2SO_4, and volatiles were removed under reduced pressure. The desired product was obtained after purification by short column chromatography on silica gel.

*Gram-Scale Preparation of **5a**.* In a Schlenk tube, 2-allenylphenylmethanamine substrates **4a** (1.5 g, 5.0 mmol) and **2a** (2.0 g, 9.9 mmol) were placed with a magnetic stir bar under a N_2 atmosphere. 1,2-Dichloroethane (50 mL, 0.1 M) and 2,6-lutidine (1.2 mL, 10.3 mmol) were added by a syringe sequentially. The resulting mixture was stirred until it became a homogeneous solution. Tf_2O (1.7 mL, 10.1 mmol) was then added by a syringe. The reaction mixture was stirred for 2 h at room temperature. Then the reaction mixture was poured into sat. $NaHCO_3$ and extracted with CH_2Cl_2. The combined organic layer was dried over anhydrous Na_2SO_4, and volatiles were removed under reduced pressure. The desired product **5a** as a white solid (2.1 g, 88% yield) was obtained after purification by short column chromatography on silica gel.

*2-(1-(Diphenylphosphanyl)vinyl)-1-tosylpyrrolidine (**3a**).* Compound **3a** was prepared according to the general procedure and was isolated as a white solid (75 mg, 86% yield) after short column chromatography (PE/EtOAc = 8:1). ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.72−7.63 (m, 2H), 7.52 (t, *J* = 10.2 Hz, 2H), 7.51−7.41 (m, 3H), 7.42−7.29 (m, 5H), 7.21 (d, *J* = 8.1 Hz, 2H), 6.10 (d, *J* = 15.3 Hz, 1H), 5.02 (d, *J* = 7.1 Hz, 1H), 4.22 (d, *J* = 8.0 Hz, 1H), 3.64−3.49 (m, 1H), 3.27−3.06 (m, 1H), 2.39 (s, 3H), 2.03−1.93 (m, 1H), 1.90−1.74 (m, 1H), 1.59−1.40 (m, 2H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 149.2 (d, *J* = 17.6 Hz), 143.4, 135.5 (d, *J* = 10.1 Hz), 135.0, 134.8, 134.6, 134.4, 134.4 (d, *J* = 20.2 Hz), 129.7, 129.4, 129.2, 128.7 (d, *J* = 22.1 Hz), 127.6, 123.4 (d, *J* = 5.3 Hz), 63.6 (d, *J* = 31.2 Hz), 49.6, 32.8 (d, *J* = 8.0 Hz), 23.6, 21.6. ^{31}P NMR (162 MHz, $CDCl_3$): δ −8.7. HRMS−ESI (*m*/*z*): $[M + H]^+$ calcd for $C_{25}H_{27}NO_2PS$, 436.1500; found, 436.1504.

*Diphenyl(1-(1-tosylpyrrolidin-2-yl)vinyl)phosphine Oxide (**3a′**).* Compound **3a′** was prepared according to the general procedure and was isolated as a white solid (65 mg, 72% yield) after short column chromatography (PE/EtOAc = 2:1). ^{1}H NMR (400 MHz, $CDCl_3$): δ 8.04−7.90 (m, 2H), 7.72−7.46 (m, 10H), 7.18 (d, *J* = 8.0 Hz, 2H), 6.42 (d, *J* = 41.4 Hz, 1H), 5.53 (d, *J* = 20.2 Hz, 1H), 4.47−4.33 (m, 1H), 3.59 (ddd, *J* = 9.6, 7.0, 2.7 Hz, 1H), 3.11 (td, *J* = 9.4, 6.5 Hz, 1H), 2.38 (s, 3H), 2.20−2.09 (m, 1H), 1.81−1.72 (m, 1H), 1.69−1.61 (m, 1H), 1.60−1.53 (m, 1H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 145.3, 144.4, 143.7, 133.8, 132.4 (d, *J* = 10.0 Hz), 132.3 (d, *J* = 2.6 Hz), 132.2 (d, *J* = 2.8 Hz), 132.1 (d, *J* = 9.7 Hz), 130.1 (d, *J* = 9.2 Hz), 129.8, 128.8 (d, *J* = 12.2 Hz), 128.6 (d, *J* = 12.2 Hz), 127.6, 61.0 (d, *J* = 18.7 Hz), 49.7, 33.3, 23.3, 21.6. ^{31}P NMR (162 MHz, $CDCl_3$): δ 31.4. HRMS−ESI (*m*/*z*): $[M + H]^+$ calcd for $C_{25}H_{27}NO_3PS$, 452.1449; found, 452.1452.

*2-(1-(Diphenylphosphanyl)vinyl)-1-(phenylsulfonyl)pyrrolidine (**3b**).* Compound **3b** was prepared according to the general procedure and was isolated as a white solid (60 mg, 71% yield) after short column chromatography (PE/EtOAc = 8:1). ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.73−7.62 (m, 4H), 7.56−7.34 (m, 11H), 6.09 (d, *J* = 15.4 Hz, 1H), 5.04 (d, *J* = 7.2 Hz, 1H), 4.25 (d, *J* = 8.1 Hz, 1H), 3.59−3.51 (m, 1H), 3.24−3.14 (m, 1H), 2.05−1.91 (m, 1H), 1.87−1.76 (m, 1H), 1.61−1.53 (m, 1H), 1.52−1.43 (m, 1H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 149.1 (d, *J* = 17.8 Hz), 137.5, 135.4 (d, *J* = 10.2 Hz), 135.0, 134.8, 134.3 (d, *J* = 20.3 Hz), 132.6, 129.5, 129.2, 129.1, 128.8 (d, *J* = 7.6 Hz), 128.6 (d, *J* = 7.5 Hz), 127.5, 123.5 (d, *J* = 5.5 Hz), 63.7 (d, *J* = 31.1 Hz), 49.5, 32.8 (d, *J* = 7.9 Hz), 23.6. ^{31}P NMR (162 MHz, $CDCl_3$): δ −8.6. HRMS−ESI (*m*/*z*): $[M + H]^+$ calcd for $C_{24}H_{25}NO_2PS$, 422.1344; found, 422.1343.

*2-(1-(Diphenylphosphanyl)vinyl)-1-((4-nitrophenyl)sulfonyl)-pyrrolidine (**3c**).* Compound **3c** was prepared according to the general procedure and was isolated as a white solid (60 mg, 65% yield) after short column chromatography (PE/EtOAc = 8:1). ^{1}H NMR (400 MHz, $CDCl_3$): δ 8.27 (d, *J* = 8.6 Hz, 2H), 7.82 (d, *J* = 8.6 Hz, 2H), 7.72−7.59 (m, 2H), 7.57−7.43 (m, 3H), 7.43−7.31 (m, 5H), 6.00 (d, *J* = 15.0 Hz, 1H), 5.06 (d, *J* = 6.9 Hz, 1H), 4.21 (d, *J* = 7.8 Hz, 1H), 3.67−3.52 (m, 1H), 3.25−3.13 (m, 1H), 2.10−1.97 (m, 1H), 1.97−1.79 (m, 1H), 1.65−1.54 (m, 2H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 150.1, 148.9 (d, *J* = 18.7 Hz), 143.5, 135.0 (d, *J* = 10.0 Hz), 134.9 (d, *J* = 20.7 Hz), 134.3 (d, *J* = 20.2 Hz), 134.1 (d, *J* = 9.2 Hz), 129.7, 129.3, 128.9 (d, *J* = 7.6 Hz), 128.7, 128.7, 124.4, 123.4 (d, *J* = 5.5 Hz), 64.0 (d, *J* = 31.4 Hz), 49.7, 33.0 (d, *J* = 7.8 Hz), 23.6. ^{31}P NMR (162 MHz, $CDCl_3$): δ −8.8. HRMS−ESI (*m*/*z*): $[M + H]^+$ calcd for $C_{24}H_{24}N_2O_4PS$, 467.1194; found, 467.1190.

*2-(1-(Diphenylphosphanyl)vinyl)-4,4-dimethyl-1-tosylpyrrolidine (**3d**).* Compound **3d** was prepared according to the general procedure and was isolated as a white solid (70 mg, 76% yield) after short column chromatography (PE/EtOAc = 8:1). ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.69 (td, *J* = 7.7, 2.1 Hz, 2H), 7.57 (d, *J* = 8.2 Hz, 2H), 7.51−7.41 (m, 3H), 7.39−7.29 (m, 5H), 7.21 (d, *J* = 8.0 Hz, 2H), 6.18 (d, *J* = 14.9 Hz, 1H), 5.03 (d, *J* = 7.2 Hz, 1H), 4.29−4.17 (m, 1H), 3.29 (d, *J* = 10.7 Hz, 1H), 3.21 (d, *J* = 10.6 Hz, 1H), 2.40 (s, 3H), 1.88−1.71 (m, 2H), 1.01 (s, 3H), 0.44 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 150.0 (d, *J* = 17.0 Hz), 143.3, 135.5 (d, *J* = 10.0 Hz), 134.9, 134.9 (d, *J* = 20.5 Hz), 134.6 (d, *J* = 9.2 Hz), 134.3 (d, *J* = 20.0 Hz), 129.6, 129.4, 129.0, 128.7 (d, *J* = 7.5 Hz), 128.5 (d, *J* = 7.3 Hz), 127.7, 122.9 (d, *J* = 3.5 Hz), 64.4 (d, *J* = 32.7 Hz), 48.8 (d, *J* = 6.6 Hz), 38.3, 26.4, 26.0, 21.6. ^{31}P NMR (162 MHz, $CDCl_3$): δ −8.8. HRMS−ESI (*m*/*z*): $[M + H]^+$ calcd for $C_{27}H_{31}NO_2PS$, 464.1813; found, 464.1816.

*3-(1-(Diphenylphosphanyl)vinyl)-2-tosyl-2-azaspiro[4.5]decane (**3e**).* Compound **3e** was prepared according to the general procedure and was isolated as a white solid (80 mg, 79% yield) after short column chromatography (PE/EtOAc = 8:1). ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.76−7.63 (m, 2H), 7.56 (d, *J* = 8.2 Hz, 2H), 7.53−7.41 (m, 3H), 7.34 (d, *J* = 3.8 Hz, 5H), 7.20 (d, *J* = 8.0 Hz, 2H), 6.17 (d, *J* = 15.1 Hz, 1H), 5.01 (d, *J* = 7.3 Hz, 1H), 4.23−4.06 (m, 1H), 3.37 (d, *J* = 11.0 Hz, 1H), 3.24 (d, *J* = 11.0 Hz, 1H), 2.39 (s, 3H), 1.85 (dd, *J* = 12.9, 8.1 Hz, 1H), 1.74 (dd, *J* = 13.0, 7.8 Hz, 1H), 1.42−1.09 (m, 10H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 150.3(d, *J* = 17.1 Hz), 143.3, 135.6 (d, *J* = 10.0 Hz), 135.0 (d, *J* = 20.4 Hz), 134.7 (d, *J* = 8.9 Hz), 134.3 (d, *J* = 19.9 Hz), 129.6, 129.4, 129.0, 128.8 (d, *J* = 7.5 Hz), 128.5 (d, *J* = 7.4 Hz), 127.7, 122.7, 63.7 (d, *J* = 32.1 Hz), 42.3, 36.3, 34.6, 26.0, 23.8, 23.1, 21.6. ^{31}P NMR (162 MHz, $CDCl_3$): δ −8.8. HRMS−ESI (*m*/*z*): $[M + H]^+$ calcd for $C_{30}H_{35}NO_2PS$, 504.2126; found, 504.2112.

*1-(1-(Diphenylphosphanyl)vinyl)-2-tosylisoindoline (**5a**).* Compound **5a** was prepared according to the general procedure and was isolated as a white solid (90 mg, 93% yield) after short column chromatography (PE/EtOAc = 8:1). ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.64 (d, *J* = 8.2 Hz, 2H), 7.36−7.14 (m, 14H), 7.08 (t, *J* = 6.7 Hz, 2H), 6.15 (d, *J* = 11.1 Hz, 1H), 5.66 (d, *J* = 11.1 Hz, 1H), 5.17 (d, *J* = 5.5 Hz, 1H), 4.64 (s, 2H), 2.37 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 151.0 (d, *J* = 19.3 Hz), 143.5, 138.9, 136.1, 135.5 (d, *J* = 9.7 Hz), 135.3, 135.2 (d, *J* = 10.1 Hz), 134.6 (d, *J* = 21.3 Hz), 134.2 (d, *J* = 19.9 Hz), 129.8, 129.0, 128.8, 128.3 (d, *J* = 5.9 Hz), 128.3, 128.2 (d, *J* = 10.6 Hz), 127.6, 127.5, 125.3, 123.9 (d, *J* = 5.5 Hz), 122.3, 71.8 (d, *J* = 29.5 Hz), 54.6, 21.6. ^{31}P NMR (162 MHz, $CDCl_3$): δ −14.4. HRMS−ESI (*m*/*z*): $[M + H]^+$ calcd for $C_{29}H_{27}NO_2PS$, 484.1500; found, 484.1505.

*1-(1-(Diphenylphosphanyl)vinyl)-5-methyl-2-tosylisoindoline (**5b**).* Compound **5b** was prepared according to the general procedure and was isolated as a white solid (90 mg, 90% yield) after short column chromatography (PE/EtOAc = 8:1). ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.62 (d, *J* = 8.2 Hz, 2H), 7.42−7.27 (m, 10H), 7.20 (d, *J* = 8.1 Hz, 2H), 7.03 (d, *J* = 7.7 Hz, 1H), 6.90 (d, *J* = 8.2 Hz, 2H), 6.13 (d, *J* = 11.2 Hz, 1H), 5.60 (d, *J* = 11.2 Hz, 1H), 5.14 (d, *J* = 5.5 Hz, 1H), 4.63−4.54 (m, 2H), 2.36 (s, 3H), 2.29 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 151.3 (d, *J* = 18.6 Hz), 143.5, 138.1, 136.2, 136.2 (d, *J* = 1.6 Hz), 135.4, 135.3 (d, *J* = 1.4 Hz), 135.3, 134.6 (d, *J* = 21.1 Hz), 134.3 (d, *J* = 20.0 Hz), 129.8, 128.8 (d, *J* = 7.1 Hz), 128.4 (d, *J* = 9.2 Hz), 128.3 (d, *J* = 2.3 Hz), 128.2, 127.5, 125.0, 123.6 (d, *J* = 5.7 Hz), 122.8, 71.6 (d, *J* = 29.6 Hz), 54.6, 21.6, 21.4. ^{31}P NMR (162 MHz, $CDCl_3$): δ −14.5. HRMS−ESI (*m*/*z*): $[M + H]^+$ calcd for $C_{30}H_{29}NO_2PS$, 498.1657; found, 498.1650.

https://dx.doi.org/10.1021/acs.joc.0c02169
J. Org. Chem. 2020, 85, 15686−15692

*1-(1-(Diphenylphosphanyl)vinyl)-5-methoxy-2-tosylisoindoline (**5c**).* Compound **5c** was prepared according to the general procedure and was isolated as a white solid (95 mg, 93% yield) after short column chromatography (PE/EtOAc = 8:1). ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.63 (d, J = 8.2 Hz, 2H), 7.39−7.26 (m, 10H), 7.21 (d, J = 8.1 Hz, 2H), 6.99 (d, J = 8.4 Hz, 1H), 6.61 (dd, J = 12.3, 3.8 Hz, 2H), 6.12 (d, J = 11.1 Hz, 1H), 5.58 (d, J = 11.4 Hz, 1H), 5.14 (d, J = 5.5 Hz, 1H), 4.60 (s, 2H), 3.75 (s, 3H), 2.37 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 159.9, 151.3 (d, J = 18.7 Hz), 143.5, 137.6, 135.5, 135.4 (d, J = 1.8 Hz), 135.3, 134.6 (d, J = 21.2 Hz), 134.2 (d, J = 19.8 Hz), 130.9 (d, J = 1.7 Hz), 129.8, 128.9, 128.7, 128.3 (d, J = 6.3 Hz), 128.3 (d, J = 7.4 Hz), 127.5, 125.0, 124.8 (d, J = 5.4 Hz), 114.1, 107.1, 71.4 (d, J = 29.8 Hz), 55.5, 54.7, 21.6. ^{31}P NMR (162 MHz, $CDCl_3$): δ −14.9. HRMS−ESI (m/z): [M + H]$^+$ calcd for $C_{30}H_{29}NO_3PS$, 514.1606; found, 514.1608.

*1-(1-(Diphenylphosphanyl)vinyl)-6-fluoro-2-tosylisoindoline (**5d**).* Compound **5d** was prepared according to the general procedure and was isolated as a white solid (85 mg, 85% yield) after short column chromatography (PE/EtOAc = 8:1). ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.64 (d, J = 8.2 Hz, 2H), 7.41−7.27 (m, 10H), 7.23 (d, J = 8.1 Hz, 2H), 7.01 (dd, J = 8.2, 4.9 Hz, 1H), 6.85 (td, J = 8.7, 2.1 Hz, 1H), 6.80 (d, J = 8.7 Hz, 1H), 6.13 (d, J = 10.7 Hz, 1H), 5.63 (d, J = 11.1 Hz, 1H), 5.18 (d, J = 5.3 Hz, 1H), 4.62 (s, 2H), 2.38 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 162.4 (C−F, ^{1}JC−F = 245.5 Hz), 150.6 (d, J = 19.8 Hz), 143.7, 141.0 (C−F, ^{3}JC−F = 1.1 Hz), 140.9 (d, J = 1.2 Hz), 135.3, 135.2 (d, J = 9.4 Hz), 134.7 (d, J = 21.6 Hz), 134.7 (d, J = 9.8 Hz), 134.1 (d, J = 19.7 Hz), 131.6 (d, J = 2.3 Hz), 129.8, 129.2, 128.8, 128.4 (d, J = 3.1 Hz), 128.3 (d, J = 3.9 Hz), 127.5, 125.5, 123.6 (C−F, ^{3}JC−F = 8.8 Hz), 115.6 (C−F, ^{2}JC−F = 23.1 Hz), 111.2 (C−F, ^{2}JC−F = 5.7 Hz), 110.9 (d, J = 5.6 Hz), 71.6 (dd, J = 29.4, 2.5 Hz), 54.1 (d, J = 3.3 Hz), 21.6. ^{31}P NMR (162 MHz, $CDCl_3$): δ −14.6. HRMS−ESI (m/z): [M + H]$^+$ calcd for $C_{29}H_{26}FNO_2PS$, 502.1406; found, 502.1404.

*1-(1-(Diphenylphosphanyl)vinyl)-6-nitro-2-tosylisoindoline (**5e**).* Compound **5e** was prepared according to the general procedure and was isolated as a white solid (75 mg, 71% yield) after short column chromatography (PE/EtOAc = 8:1). ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.96 (dd, J = 8.4, 1.9 Hz, 1H), 7.80−7.64 (m, 3H), 7.33−7.17 (m, 11H), 7.12 (td, J = 7.7, 1.7 Hz, 2H), 6.17 (d, J = 10.1 Hz, 1H), 5.77 (d, J = 12.6 Hz, 1H), 5.24 (d, J = 5.1 Hz, 1H), 4.84 (s, 2H), 2.41 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 149.9 (d, J = 20.5 Hz), 147.6, 144.0, 143.4, 140.1, 135.7, 134.9 (d, J = 8.9 Hz), 134.7 (d, J = 22.2 Hz), 134.1 (d, J = 9.2 Hz), 133.7 (d, J = 18.9 Hz), 130.0, 129.5, 128.7, 128.5 (d, J = 6.7 Hz), 128.4 (d, J = 5.2 Hz), 127.5, 126.8, 123.8, 123.1, 120.0 (d, J = 3.7 Hz), 71.6 (d, J = 29.4 Hz), 54.4 (d, J = 4.8 Hz), 21.7. ^{31}P NMR (162 MHz, $CDCl_3$): δ −15.9. HRMS−ESI (m/z): [M + H]$^+$ calcd for $C_{29}H_{26}N_2O_4PS$, 529.1351; found, 529.1350.

*(1R,3S)-1-(1-(Diphenylphosphanyl)vinyl)-3-methyl-2-tosylisoindoline (**5f**).* Compound **5f** was prepared according to the general procedure and was isolated as a white solid (85 mg, 85% yield) after short column chromatography (PE/EtOAc = 8:1). ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.56 (d, J = 8.2 Hz, 2H), 7.49−7.39 (m, 4H), 7.35 (dd, J = 7.8, 4.8 Hz, 6H), 7.27−7.13 (m, 5H), 7.07 (d, J = 7.5 Hz, 1H), 6.23 (d, J = 10.8 Hz, 1H), 5.51 (d, J = 10.4 Hz, 1H), 5.18 (d, J = 5.4 Hz, 1H), 4.91 (q, J = 6.3 Hz, 1H), 2.35 (s, 3H), 1.68 (d, J = 6.5 Hz, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 150.5 (d, J = 19.2 Hz), 143.6, 141.0, 138.0, 135.5 (d, J = 10.3 Hz), 135.4 (d, J = 9.4 Hz), 134.8, 134.8 (d, J = 21.0 Hz), 134.4 (d, J = 19.9 Hz), 129.8, 129.0, 128.9, 128.5 (d, J = 4.3 Hz), 128.4 (d, J = 4.0 Hz), 127.8, 127.6, 126.1, 124.0 (d, J = 8.7 Hz), 122.3, 71.8 (d, J = 31.8 Hz), 62.6, 24.0 (d, J = 3.7 Hz), 21.6. ^{31}P NMR (162 MHz, $CDCl_3$): δ −14.6. HRMS−ESI (m/z): [M + H]$^+$ calcd for $C_{30}H_{29}NO_2PS$, 498.1657; found, 498.1659.

*(1R,3S)-1-(1-(Diphenylphosphanyl)vinyl)-3-ethyl-2-tosylisoindoline (**5g**).* Compound **5g** was prepared according to the general procedure and was isolated as a white solid (90 mg, 88% yield) after short column chromatography (PE/EtOAc = 8:1). ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.83 (d, J = 8.2 Hz, 2H), 7.32−7.21 (m, 10H), 7.16 (t, J = 7.1 Hz, 2H), 7.05 (t, J = 7.3 Hz, 1H), 6.99 (d, J = 7.6 Hz, 1H), 6.74 (t, J = 7.4 Hz, 1H), 6.67 (d, J = 7.6 Hz, 1H), 6.05 (d, J = 10.2 Hz, 1H), 5.67 (dd, J = 12.7, 2.3 Hz, 1H), 5.51 (s, 1H), 5.37 (d, J = 5.3 Hz, 1H), 2.42 (s, 3H), 2.41−2.36 (m, 1H), 1.92−1.75 (m, 1H), 0.13 (t, J = 7.3 Hz, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 152.5 (d, J = 21.3 Hz), 142.9, 139.3 (d, J = 9.8 Hz), 137.5 (d, J = 1.7 Hz), 136.9 (d, J = 10.8 Hz), 135.8 (d, J = 10.8 Hz), 134.8 (d, J = 22.7 Hz), 133.8 (d, J = 17.9 Hz), 129.5, 128.9, 128.1 (d, J = 3.9 Hz), 128.0 (d, J = 2.6 Hz), 128.0 (d, J = 6.3 Hz), 127.2, 127.2, 125.4 (d, J = 1.6 Hz), 124.1 (d, J = 1.9 Hz), 121.8, 73.2 (d, J = 31.4 Hz), 67.9 (d, J = 4.5 Hz), 26.5, 21.7, 7.0. ^{31}P NMR (162 MHz, $CDCl_3$): δ −15.7. HRMS−ESI (m/z): [M + H]$^+$ calcd for $C_{31}H_{31}NO_2PS$, 512.1813; found, 512.1816.

*1-(1-(Diphenylphosphanyl)vinyl)-1-methyl-2-tosylisoindoline (**5h**).* Compound **5h** was prepared according to the general procedure and was isolated as a white solid (80 mg, 80% yield) after short column chromatography (PE/EtOAc = 8:1). ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.78 (d, J = 8.3 Hz, 2H), 7.35−7.24 (m, 8H), 7.20−7.15 (m, 4H), 7.13−7.07 (m, 2H), 6.97−6.90 (m, 1H), 6.84 (d, J = 7.8 Hz, 1H), 6.27 (d, J = 9.3 Hz, 1H), 5.36 (d, J = 5.2 Hz, 1H), 4.89 (d, J = 12.6 Hz, 1H), 4.59 (d, J = 12.6 Hz, 1H), 2.40 (s, 3H), 1.98 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 155.3 (d, J = 19.6 Hz), 143.7, 143.2, 138.2, 137.3 (d, J = 10.6 Hz), 136.5 (d, J = 11.1 Hz), 134.9 (d, J = 1.2 Hz), 134.2 (d, J = 21.8 Hz), 133.7 (d, J = 18.7 Hz), 129.6, 128.7, 128.3, 128.2 (d, J = 2.9 Hz), 128.1 (d, J = 1.7 Hz), 128.0, 127.7, 127.5, 124.7 (d, J = 2.7 Hz), 123.2 (d, J = 3.3 Hz), 122.2, 54.6 (d, J = 6.3 Hz), 29.9, 25.9 (d, J = 5.1 Hz), 21.6. ^{31}P NMR (162 MHz, $CDCl_3$): δ −13.5. HRMS−ESI (m/z): [M + H]$^+$ calcd for $C_{30}H_{29}NO_2PS$, 498.1657; found, 498.1677.

*(E)-1-(1-(Diphenylphosphanyl)prop-1-en-1-yl)-2-tosylisoindoline (**5i**).* Compound **5i** was prepared according to the general procedure and was isolated as a colorless oil (85 mg, 85% yield) after short column chromatography (PE/EtOAc = 8:1). ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.66 (d, J = 8.3 Hz, 2H), 7.40−7.31 (m, 5H), 7.24−6.98 (m, 10H), 6.81 (d, J = 7.6 Hz, 1H), 6.23 (d, J = 16.8 Hz, 1H), 5.79 (p, J = 6.8 Hz, 1H), 4.82−4.54 (m, 2H), 2.37 (s, 3H), 2.08 (d, J = 7.0 Hz, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 143.4, 141.9 (d, J = 14.0 Hz), 139.1 (d, J = 3.3 Hz), 137.4 (d, J = 11.2 Hz), 137.1 (d, J = 2.5 Hz), 136.7 (d, J = 1.3 Hz), 136.5 (d, J = 9.6 Hz), 135.6, 134.0, 133.8 (d, J = 2.1 Hz), 133.5, 129.7, 128.2, 128.1, 128.0 (d, J = 11.5 Hz), 127.9 (d, J = 7.2 Hz), 127.5, 127.4, 123.4, 122.2, 65.6 (d, J = 31.8 Hz), 54.8 (d, J = 5.3 Hz), 21.5, 15.4. ^{31}P NMR (162 MHz, $CDCl_3$): δ −13.9. HRMS−ESI (m/z): [M + H]$^+$ calcd for $C_{30}H_{29}NO_3PS$, 498.1657; found, 498.1659.

*(E)-1-(1-(Diphenylphosphanyl)hept-1-en-1-yl)-2-tosylisoindoline (**5j**).* Compound **5j** was prepared according to the general procedure and was isolated as a white solid (90 mg, 81% yield) after short column chromatography (PE/EtOAc = 8:1). ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.68 (d, J = 8.2 Hz, 2H), 7.40−7.31 (m, 5H), 7.24 (d, J = 8.1 Hz, 2H), 7.20−7.09 (m, 5H), 7.06−6.98 (m, 3H), 6.80 (d, J = 7.6 Hz, 1H), 6.21 (d, J = 17.0 Hz, 1H), 5.73 (q, J = 7.2 Hz, 1H), 4.72 (q, J = 13.8 Hz, 2H), 2.61−2.43 (m, 2H), 2.39 (s, 3H), 1.58−1.43 (m, 2H), 1.40−1.31 (m, 4H), 0.93 (t, J = 6.8 Hz, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 143.6 (d, J = 2.7 Hz), 143.5, 140.7 (d, J = 14.3 Hz), 139.2 (d, J = 3.3 Hz), 137.7 (d, J = 11.3 Hz), 136.8 (d, J = 1.4 Hz), 136.7 (d, J = 9.6 Hz), 135.8, 134.0, 133.8 (d, J = 0.8 Hz), 133.6, 129.8, 128.3 (d, J = 4.6 Hz), 128.2 (d, J = 6.7 Hz), 128.1, 127.9 (d, J = 7.2 Hz), 127.6 (d, J = 8.3 Hz), 123.5, 122.4, 66.1 (d, J = 32.2 Hz), 55.0 (d, J = 5.4 Hz), 31.8, 29.5 (d, J = 1.3 Hz), 29.3, 22.6, 21.6, 14.2. ^{31}P NMR (162 MHz, $CDCl_3$): δ −14.1. HRMS−ESI (m/z): [M + H]$^+$ calcd for $C_{34}H_{37}NO_2PS$, 554.2283; found, 554.2289.

*1-(2-Methylprop-1-en-1-yl)-2-tosylisoindoline (**5k**).* Compound **5k** was prepared according to the general procedure and was isolated as a white solid (57 mg, 87% yield) after short column chromatography (PE/EtOAc = 10/1). ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.75 (d, J = 8.2 Hz, 2H), 7.29 (d, J = 8.1 Hz, 2H), 7.27−7.20 (m, 2H), 7.20−7.13 (m, 1H), 7.06−6.97 (m, 1H), 5.63 (d, J = 9.5 Hz, 1H), 5.09 (d, J = 9.5 Hz, 1H), 4.71 (s, 2H), 2.41 (s, 3H), 1.94 (s, 3H), 1.74 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 143.4, 140.4, 135.8, 135.2, 134.6, 129.6, 127.9, 127.8, 127.7, 125.3,

https://dx.doi.org/10.1021/acs.joc.0c02169
J. Org. Chem. 2020, 85, 15686−15692

123.3, 122.5, 63.9, 53.6, 25.9, 21.6, 18.4. HRMS−ESI (m/z): [M + H]$^+$ calcd for $C_{19}H_{22}NO_2S$, 328.1371; found, 328.1379.

*1-(1-(Diphenylphosphanyl)vinyl)-2-tosyl-1,2,3,4-tetrahydroisoquinoline (**5l**).* Compound **5l** was prepared according to the general procedure and was isolated as a white solid (90 mg, 90% yield) after short column chromatography (PE/EtOAc = 8:1). ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.57 (d, J = 8.2 Hz, 2H), 7.49−7.42 (m, 2H), 7.39 (dd, J = 4.5, 1.7 Hz, 3H), 7.33 (d, J = 3.4 Hz, 5H), 7.12−7.02 (m, 4H), 6.93−6.86 (m, 2H), 5.68 (d, J = 5.0 Hz, 1H), 5.52 (d, J = 10.7 Hz, 1H), 5.12 (d, J = 4.9 Hz, 1H), 3.58 (t, J = 6.5 Hz, 2H), 2.61 (t, J = 6.3 Hz, 2H), 2.33 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 150.8 (d, J = 21.5 Hz), 143.1, 137.2, 135.7 (d, J = 12.0 Hz), 134.9 (d, J = 10.9 Hz), 134.7, 134.5 (d, J = 6.0 Hz), 134.3, 134.2, 133.7 (d, J = 4.6 Hz), 129.3, 129.2, 129.0, 128.7 (d, J = 5.8 Hz), 128.6 (d, J = 3.4 Hz), 128.5, 128.2 (d, J = 1.4 Hz), 127.5, 127.4 (d, J = 4.4 Hz), 125.9, 60.7 (d, J = 29.1 Hz), 39.9, 26.5, 21.6. ^{31}P NMR (162 MHz, $CDCl_3$): δ −10.3. HRMS−ESI (m/z): [M + H]$^+$ calcd for $C_{30}H_{29}NO_2PS$, 498.1657; found, 498.1653.

*1-(1-(Di-p-tolylphosphanyl)vinyl)-2-tosylisoindoline (**6a**).* Compound **6a** was prepared according to the general procedure and was isolated as a colorless oil (70 mg, 69% yield) after short column chromatography (PE/EtOAc = 8:1). ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.59 (d, J = 8.1 Hz, 2H), 7.28−7.01 (m, 14H), 6.08 (d, J = 11.2 Hz, 1H), 5.61 (d, J = 10.2 Hz, 1H), 5.11 (d, J = 5.5 Hz, 1H), 4.60 (s, 2H), 2.34 (s, 3H), 2.33 (s, 3H), 2.32 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 151.3 (d, J = 19.5 Hz), 143.5, 139.3 (d, J = 1.0 Hz), 138.8 (d, J = 23.9 Hz), 136.1, 135.4, 134.7 (d, J = 21.6 Hz), 134.3 (d, J = 20.1 Hz), 132.1 (d, J = 8.3 Hz), 131.8 (d, J = 8.7 Hz), 129.8, 129.2, 129.1, 129.1, 128.0, 127.6, 127.4, 124.6, 124.0 (d, J = 6.7 Hz), 122.3, 71.5 (d, J = 29.1 Hz), 54.6 (d, J = 2.6 Hz), 21.6, 21.4 (d, J = 1.1 Hz). ^{31}P NMR (162 MHz, $CDCl_3$): δ −15.6. HRMS−ESI (m/z): [M + H]$^+$ calcd for $C_{31}H_{31}NO_2PS$, 512.1813; found, 512.1824.

*1-(1-(Di([1,1′-biphenyl]-4-yl)phosphanyl)vinyl)-2-tosylisoindoline (**6b**).* Compound **6b** was prepared according to the general procedure and was isolated as a colorless oil (100 mg, 79% yield) after short column chromatography (PE/EtOAc = 8:1). ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.66 (d, J = 7.8 Hz, 2H), 7.61−7.54 (m, 6H), 7.49−7.39 (m, 10H), 7.38−7.31 (m, 2H), 7.19 (d, J = 8.1 Hz, 2H), 7.15 (d, J = 8.2 Hz, 2H), 7.06 (t, J = 8.5 Hz, 2H), 6.17 (d, J = 11.3 Hz, 1H), 5.68 (d, J = 10.8 Hz, 1H), 5.27 (d, J = 5.5 Hz, 1H), 4.64 (s, 2H), 2.34 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 151.0 (d, J = 19.5 Hz), 143.4, 141.5 (d, J = 30.9 Hz), 140.6 (d, J = 4.6 Hz), 138.8, 136.1, 135.4, 135.1 (d, J = 21.7 Hz), 134.5 (d, J = 19.8 Hz), 134.5, 134.0 (d, J = 10.2 Hz), 129.8, 129.7, 129.0 (d, J = 2.4 Hz), 128.8 (d, J = 4.4 Hz), 128.2, 128.0, 127.6, 127.6, 127.5, 127.4, 127.3 (d, J = 4.3 Hz), 127.1 (d, J = 3.7 Hz), 126.9 (d, J = 7.7 Hz), 126.9 (d, J = 8.6 Hz), 125.2, 123.9 (d, J = 5.4 Hz), 122.23, 71.7 (d, J = 29.0 Hz), 54.57, 21.5. ^{31}P NMR (162 MHz, $CDCl_3$): δ −15.6. HRMS−ESI (m/z): [M + H]$^+$ calcd for $C_{41}H_{35}NO_2PS$, 636.2126; found, 636.2147.

*1-(1-(Di(naphthalen-1-yl)phosphanyl)vinyl)-2-tosylisoindoline (**6c**).* Compound **6c** was prepared according to the general procedure and was isolated as a colorless oil (108 mg, 92% yield) after short column chromatography (PE/EtOAc = 8:1). ^{1}H NMR (400 MHz, $CDCl_3$): δ 8.35 (dd, J = 8.0, 4.8 Hz, 1H), 8.11−8.01 (m, 1H), 7.89−7.76 (m, 4H), 7.61 (t, J = 5.9 Hz, 1H), 7.54 (d, J = 7.3 Hz, 2H), 7.48−7.27 (m, 8H), 7.25−7.18 (m, 2H), 7.06 (t, J = 7.4 Hz, 3H), 6.20 (d, J = 11.2 Hz, 1H), 5.82 (d, J = 11.7 Hz, 1H), 4.92 (d, J = 5.5 Hz, 1H), 4.70−4.45 (m, 2H), 2.27 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 149.0 (d, J = 16.8 Hz), 143.4, 139.3, 136.3, 135.8 (d, J = 7.4 Hz), 135.5 (d, J = 7.5 Hz), 135.0, 134.0 (d, J = 19.4 Hz), 133.5, 133.4 (d, J = 0.9 Hz), 131.5 (d, J = 11.0 Hz), 131.4 (d, J = 10.2 Hz), 129.7, 129.6, 128.5 (d, J = 1.6 Hz), 128.5 (d, J = 1.5 Hz), 128.1, 127.4, 127.3, 126.5 (d, J = 2.1 Hz), 126.3 (d, J = 7.4 Hz), 126.1 (d, J = 2.1 Hz), 126.0, 125.9 (d, J = 2.2 Hz), 125.8 (d, J = 0.8 Hz), 125.8, 125.6, 123.9 (d, J = 6.8 Hz), 122.4, 71.9 (d, J = 31.8 Hz), 54.7 (d, J = 3.5 Hz), 21.4 (s). ^{31}P NMR (162 MHz, $CDCl_3$): δ −35.1. HRMS−ESI (m/z): [M + H]$^+$ calcd for $C_{37}H_{31}NO_2PS$, 584.1813; found, 584.1788.

*1-(1-(Di(naphthalen-2-yl)phosphanyl)vinyl)-2-tosylisoindoline (**6d**).* Compound **6d** was prepared according to the general procedure and was isolated as a colorless oil (90 mg, 77% yield) after short column chromatography (PE/EtOAc = 8:1). ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.87−7.70 (m, 8H), 7.59 (d, J = 7.5 Hz, 2H), 7.52−7.40 (m, 6H), 7.16 (d, J = 7.6 Hz, 1H), 7.12−7.05 (m, 3H), 7.04−6.93 (m, 2H), 6.20 (d, J = 11.3 Hz, 1H), 5.71 (d, J = 10.8 Hz, 1H), 5.27 (d, J = 5.5 Hz, 1H), 4.64 (s, 2H), 2.29 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 150.6 (d, J = 19.4 Hz), 143.4, 138.8, 136.0, 135.5, 135.2 (d, J = 5.8 Hz), 134.5 (d, J = 22.3 Hz), 133.4 (d, J = 13.4 Hz), 133.3, 133.2, 133.1 (d, J = 4.4 Hz), 133.0, 132.6 (d, J = 10.3 Hz), 130.5 (d, J = 16.9 Hz), 130.3 (d, J = 17.2 Hz), 129.6, 128.2, 128.1 (d, J = 6.2 Hz), 127.6 (d, J = 6.8 Hz), 127.6 (d, J = 7.3 Hz), 127.4, 127.3, 126.7 (d, J = 15.4 Hz), 126.1, 125.5, 123.9 (d, J = 6.0 Hz), 122.2, 71.7 (d, J = 29.2 Hz), 54.6 (d, J = 2.8 Hz), 21.4. ^{31}P NMR (162 MHz, $CDCl_3$): δ −12.9. HRMS−ESI (m/z): [M + H]$^+$ calcd for $C_{37}H_{31}NO_2PS$, 584.1813; found, 584.1810.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.0c02169.

Calculation results; copies of NMR spectra; X-ray crystallographic data for **5g** (PDF)

Accession Codes

CCDC 2024402 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

■ AUTHOR INFORMATION

Corresponding Authors

Chun-Hua Yang − *Henan Province Key Laboratory of New Optoelectronic Functional Materials, College of Chemistry and Chemical Engineering, Anyang Normal University, Anyang 455000, People's Republic of China;* orcid.org/0000-0002-3770-9887; Email: 09yangchunhua@163.com

Junbiao Chang − *School of Pharmaceutical Sciences, Zhengzhou University, Zhengzhou, Henan 450001, People's Republic of China;* orcid.org/0000-0001-6236-1256; Email: changjunbiao@zzu.edu.cn

Authors

Zhantao Yang − *Henan Province Key Laboratory of New Optoelectronic Functional Materials, College of Chemistry and Chemical Engineering, Anyang Normal University, Anyang 455000, People's Republic of China; School of Pharmaceutical Sciences, Zhengzhou University, Zhengzhou, Henan 450001, People's Republic of China;* orcid.org/0000-0003-1575-3665

Zhenzhen Sun − *Henan Province Key Laboratory of New Optoelectronic Functional Materials, College of Chemistry and Chemical Engineering, Anyang Normal University, Anyang 455000, People's Republic of China*

Ningning Zhu − *Henan Province Key Laboratory of New Optoelectronic Functional Materials, College of Chemistry and Chemical Engineering, Anyang Normal University, Anyang 455000, People's Republic of China*

Wenyan Li − *Henan Province Key Laboratory of New Optoelectronic Functional Materials, College of Chemistry and Chemical Engineering, Anyang Normal University, Anyang 455000, People's Republic of China*

Meng Han − *Henan Province Key Laboratory of New Optoelectronic Functional Materials, College of Chemistry and*

Chemical Engineering, Anyang Normal University, Anyang 455000, People's Republic of China

Yan Qiao − *School of Basic Medicine, Zhengzhou University, Zhengzhou, Henan 450001, People's Republic of China;* orcid.org/0000-0001-6357-8203

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.joc.0c02169

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We acknowledge financial support from the National Natural Science Foundation of China (21801006 and 81703353), China Postdoctoral Science Foundation (2018T110744), Natural Science Foundation of Henan Province (182300410295), the Industrial Research Project of Technology Bureau of Anyang and the Program for Innovative Research Team of Science and Technology in the University of Henan Province (S202010479009 and X202010479090). We also thank Prof. Hankui Wu for the helpful discussions.

■ REFERENCES

(1) (a) Dutartre, M.; Bayardon, J.; Jugé, S. Applications and stereoselective syntheses of P-chirogenic phosphorus compounds. *Chem. Soc. Rev.* **2016**, *45*, 5771−5794. (b) Queffélec, C.; Petit, M.; Janvier, P.; Knight, D. A.; Bujoli, B. Surface Modification Using Phosphonic Acids and Esters. *Chem. Rev.* **2012**, *112*, 3777−3807.

(2) (a) Erion, M. D.; Dang, Q.; Reddy, M. R.; Kasibhatla, S. R.; Huang, J.; Lipscomb, W. N.; van Poelje, P. D. Structure-Guided Design of AMP Mimics That Inhibit Fructose-1,6-bisphosphatase with High Affinity and Specificity. *J. Am. Chem. Soc.* **2007**, *129*, 15480−15490. (b) Dang, Q.; Brown, B. S.; Liu, Y.; Rydzewski, R. M.; Robinson, E. D.; Poelje, P. D. v.; Reddy, M. R.; Erion, M. D. Fructose-1,6-bisphosphatase Inhibitors. 1. Purine Phosphonic Acids as Novel AMP Mimics. *J. Med. Chem.* **2009**, *52*, 2880−2898.

(3) (a) McManus, H. A.; Guiry, P. J. Recent Developments in the Application of Oxazoline-Containing Ligands in Asymmetric Catalysis. *Chem. Rev.* **2004**, *104*, 4151−4202. (b) Tang, W.; Zhang, X. New Chiral Phosphorus Ligands for Enantioselective Hydrogenation. *Chem. Rev.* **2003**, *103*, 3029−3070. (c) Xie, J.-H.; Zhou, Q.-L. Chiral Diphosphine and Monodentate Phosphorus Ligands on a Spiro Scaffold for Transition-Metal-Catalyzed Asymmetric Reactions. *Acc. Chem. Res.* **2008**, *41*, 581−593. (d) Pereira, M. M.; Calvete, M. J. F.; Carrilho, R. M. B.; Abreu, A. R. Synthesis of binaphthyl based phosphine and phosphite ligands. *Chem. Soc. Rev.* **2013**, *42*, 6990−7027.

(4) (a) Guo, R.; Morris, R. H.; Song, D. Enantioselective Tandem Michael Addition/H2-Hydrogenation Catalyzed by Ruthenium Hydride Borohydride Complexes Containing β-aminophosphine Ligands. *J. Am. Chem. Soc.* **2005**, *127*, 516−517. (b) Li, W.; Zhang, J. Recent developments in the synthesis and utilization of chiral β-aminophosphine derivatives as catalysts or ligands. *Chem. Soc. Rev.* **2016**, *45*, 1657−1677. (c) Tolman, C. A. Steric effects of phosphorus ligands in organometallic chemistry and homogeneous catalysis. *Chem. Rev.* **1977**, *77*, 313−348.

(5) Cheviet, T.; Wein, S.; Bourchenin, G.; Lagacherie, M.; Périgaud, C.; Cerdan, R.; Peyrottes, S. β-Hydroxy- and β-Aminophosphonate Acyclonucleosides as Potent Inhibitors of Plasmodium falciparum Growth. *J. Med. Chem.* **2020**, *63*, 8069−8087.

(6) Yokomatsu, T.; Sato, M.; Shibuya, S. Lipase-catalyzed enantioselective acylation of prochiral 2-(ω-phosphono)alkyl-1,3-propanediols: Application to the enantioselective synthesis of ω-phosphono-α-amino acids. *Tetrahedron: Asymmetry* **1996**, *7*, 2743−2754.

(7) Palacios, F.; Alonso, C.; de los Santos, J. M. Synthesis of β-Aminophosphonates and -Phosphinates. *Chem. Rev.* **2005**, *105*, 899−932.

(8) (a) Okugawa, Y.; Hirano, K.; Miura, M. Copper-Catalyzed Vicinal Diphosphination of Styrenes: Access to 1,2-Bis(diphenylphosphino)ethane-Type Bidentate Ligands from Olefins. *Angew. Chem., Int. Ed.* **2016**, *55*, 13558−13561. (b) Yuan, J.; Zhu, L.; Zhang, J.; Li, J.; Cui, C. Sequential Addition of Phosphine to Alkynes for the Selective Synthesis of 1,2-Diphosphinoethanes under Catalysis. Well-Defined NHC-Copper Phosphides vs in Situ CuCl2/NHC Catalyst. *Organometallics* **2017**, *36*, 455−459. (c) Di Giuseppe, A.; De Luca, R.; Castarlenas, R.; Pérez-Torrente, J. J.; Crucianelli, M.; Oro, L. A. Double hydrophosphination of alkynes promoted by rhodium: the key role of an N-heterocyclic carbene ligand. *Chem. Commun.* **2016**, *52*, 5554−5557. (d) Kamitani, M.; Itazaki, M.; Tamiya, C.; Nakazawa, H. Regioselective Double Hydrophosphination of Terminal Arylacetylenes Catalyzed by an Iron Complex. *J. Am. Chem. Soc.* **2012**, *134*, 11932−11935.

(9) Zhang, H.-Y.; Mao, L.-L.; Yang, B.; Yang, S.-D. Copper-catalyzed radical cascade cyclization for the synthesis of phosphorated indolines. *Chem. Commun.* **2015**, *51*, 4101−4104.

(10) Li, Y.-H.; Wang, C.-H.; Gao, S.-Q.; Qi, F.-M.; Yang, S.-D. Visible photocatalysis of novel oxime phosphonates: synthesis of β-aminophosphonates. *Chem. Commun.* **2019**, *55*, 11888−11891.

(11) Li, J.-A.; Zhang, P.-Z.; Liu, K.; Shoberu, A.; Zou, J.-P.; Zhang, W. Phosphinoyl Radical-Initiated α,β-Aminophosphinoylation of Alkenes. *Org. Lett.* **2017**, *19*, 4704−4706.

(12) Wang, Y.; Wang, W.; Tang, R.; Liu, Z.; Tao, W.; Fang, Z. Iron(iii)-catalyzed radical α,β-aminophosphinoylation of styrenes with diphenylphosphine oxides and anilines. *Org. Biomol. Chem.* **2018**, *16*, 7782−7786.

(13) Xu, J.; Li, X.; Gao, Y.; Zhang, L.; Chen, W.; Fang, H.; Tang, G.; Zhao, Y. Mn(iii)-mediated phosphonation−azidation of alkenes: a facile synthesis of β-azidophosphonates. *Chem. Commun.* **2015**, *51*, 11240−11243.

(14) (a) Sun, C.-L.; Shi, Z.-J. Transition-Metal-Free Coupling Reactions. *Chem. Rev.* **2014**, *114*, 9219−9280. (b) Roscales, S.; Csákÿ, A. G. Transition-metal-free C−C bond forming reactions of aryl, alkenyl and alkynylboronic acids and their derivatives. *Chem. Soc. Rev.* **2014**, *43*, 8215−8225. (c) Zhu, C.; Falck, J. R. Transition Metal-Free ipso-Functionalization of Arylboronic Acids and Derivatives. *Adv. Synth. Catal.* **2014**, *356*, 2395−2410.

(15) (a) Unoh, Y.; Hirano, K.; Miura, M. Metal-Free Electrophilic Phosphination/Cyclization of Alkynes. *J. Am. Chem. Soc.* **2017**, *139*, 6106−6109. (b) Nishimura, K.; Hirano, K.; Miura, M. Synthesis of Dibenzophospholes by Tf2O-Mediated Intramolecular Phospha-Friedel−Crafts-Type Reaction. *Org. Lett.* **2019**, *21*, 1467−1470. (c) Nishimura, K.; Unoh, Y.; Hirano, K.; Miura, M. Phosphenium-Cation-Mediated Formal Cycloaddition Approach to Benzophospholes. *Chem. - Eur. J.* **2018**, *24*, 13089−13092. (d) Nishimura, K.; Hirano, K.; Miura, M. Direct Synthesis of Dibenzophospholes from Biaryls by Double C−P Bond Formation via Phosphenium Dication Equivalents. *Org. Lett.* **2020**, *22*, 3185−3189.

(16) Ohmiya, H.; Yorimitsu, H.; Oshima, K. Cobalt-Catalyzed syn Hydrophosphination of Alkynes. *Angew. Chem., Int. Ed.* **2005**, *44*, 2368−2370.

(17) Yang, C.-H.; Fan, H.; Li, H.; Hou, S.; Sun, X.; Luo, D.; Zhang, Y.; Yang, Z.; Chang, J. Direct Access to Allenylphosphine Oxides via a Metal Free Coupling of Propargylic Substrates with P(O)H Compounds. *Org. Lett.* **2019**, *21*, 9438−9441.

(18) Yang, C.-H.; Han, M.; Li, W.; Zhu, N.; Sun, Z.; Wang, J.; Yang, Z.; Li, Y.-M. Direct Intramolecular Aminoboration of Allenes. *Org. Lett.* **2020**, *22*, 5090−5093.